In [56]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.decomposition import PCA
```

In [57]:
```
df=pd.read_csv('mnist_test.csv')
```

In [58]:
```
df.head()
```

Out[58]:

	label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8	1x9	...	28x19	28x20	28x21	28x22
0	7	0	0	0	0	0	0	0	0	0	...	0	0	0	0
1	2	0	0	0	0	0	0	0	0	0	...	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
4	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0

5 rows × 785 columns

In [59]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Columns: 785 entries, label to 28x28
dtypes: int64(785)
memory usage: 59.9 MB
```

In [60]:
```
df.describe()
```

Out[60]:

	label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8
count	10000.000000	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0
mean	4.443400	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
std	2.895865	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
min	0.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	2.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
50%	4.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
75%	7.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
max	9.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

8 rows × 785 columns

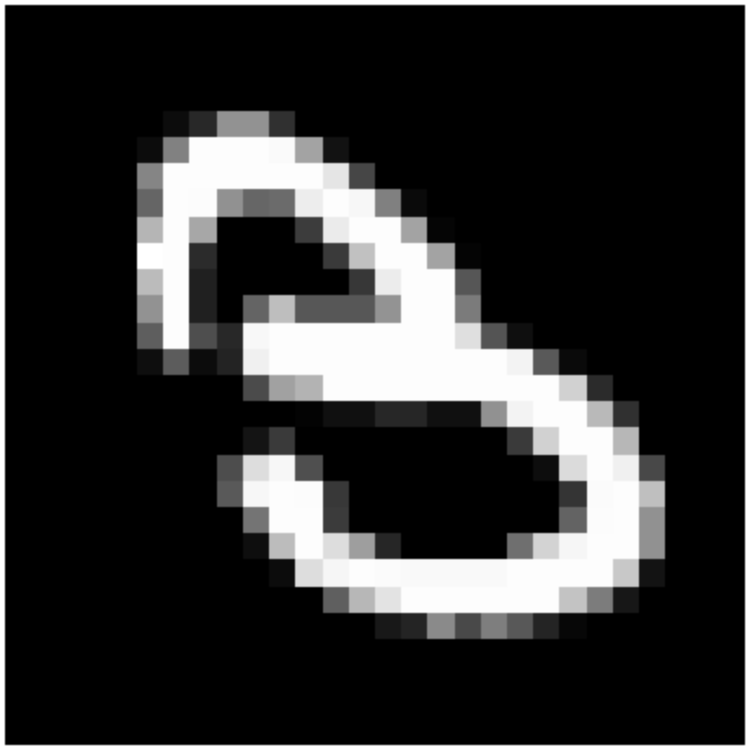
In [61]:
```
X = df.drop('label', axis=1)
y = df['label']
```

In [ ]:

In [62]:
```
model = LogisticRegression(max_iter=1000)
model.fit(X, y)
```

Out[62]:
```
▾ LogisticRegression
▶ Parameters
```

In [63]:
```
img = X[df['label'] ==3 ].iloc[0].values
plt.imshow(img.reshape(28,28), cmap='gray')
plt.axis('off')
plt.show()
```

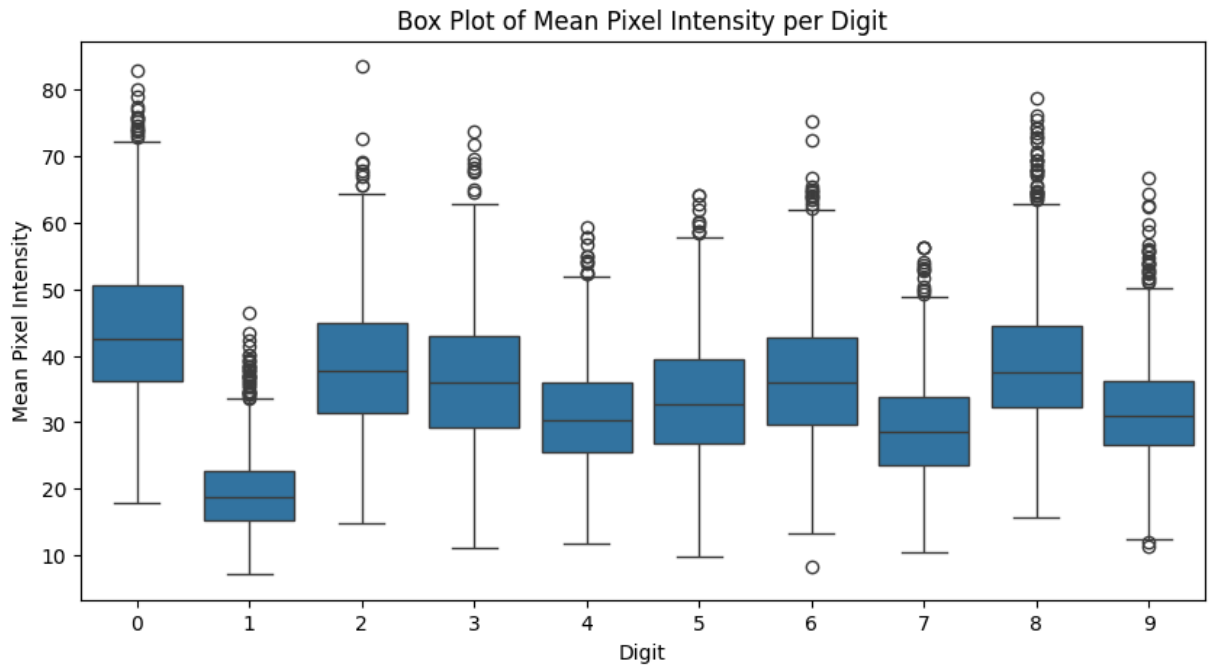


In [ ]:

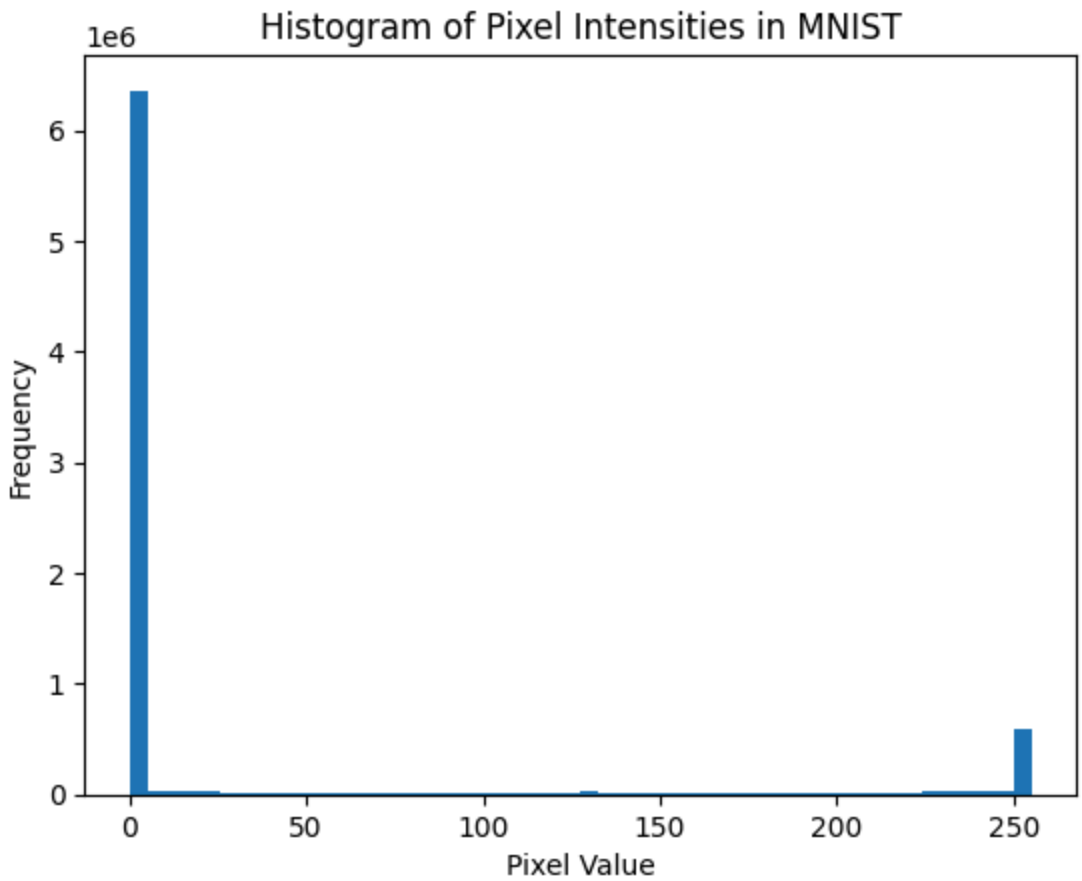
In [64]:
```
X.shape   # (n_samples, 784)
y.unique()
```

Out[64]:
```
array([7, 2, 1, 0, 4, 9, 5, 6, 3, 8])
```

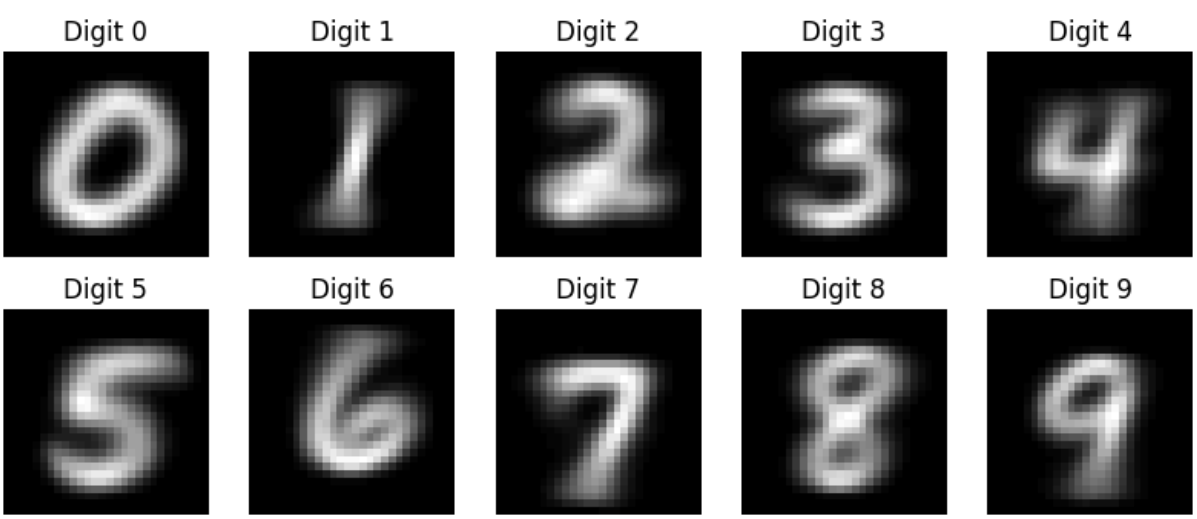
In [74]:
```
import pandas as pd
import seaborn as sns

# Compute mean pixel intensity per image
mean_pixel_intensity = X.mean(axis=1)

box_df = pd.DataFrame({
    "Digit": y,
    "Mean Pixel Intensity": mean_pixel_intensity
})

plt.figure(figsize=(10,5))
sns.boxplot(
    x="Digit",
    y="Mean Pixel Intensity",
    data=box_df
)
plt.title("Box Plot of Mean Pixel Intensity per Digit")
plt.show()
```



In [73]:
```
import matplotlib.pyplot as plt

# Histogram of all pixel values
plt.hist(X.values.flatten(), bins=50)
plt.title("Histogram of Pixel Intensities in MNIST")
plt.xlabel("Pixel Value")
plt.ylabel("Frequency")
plt.show()
```

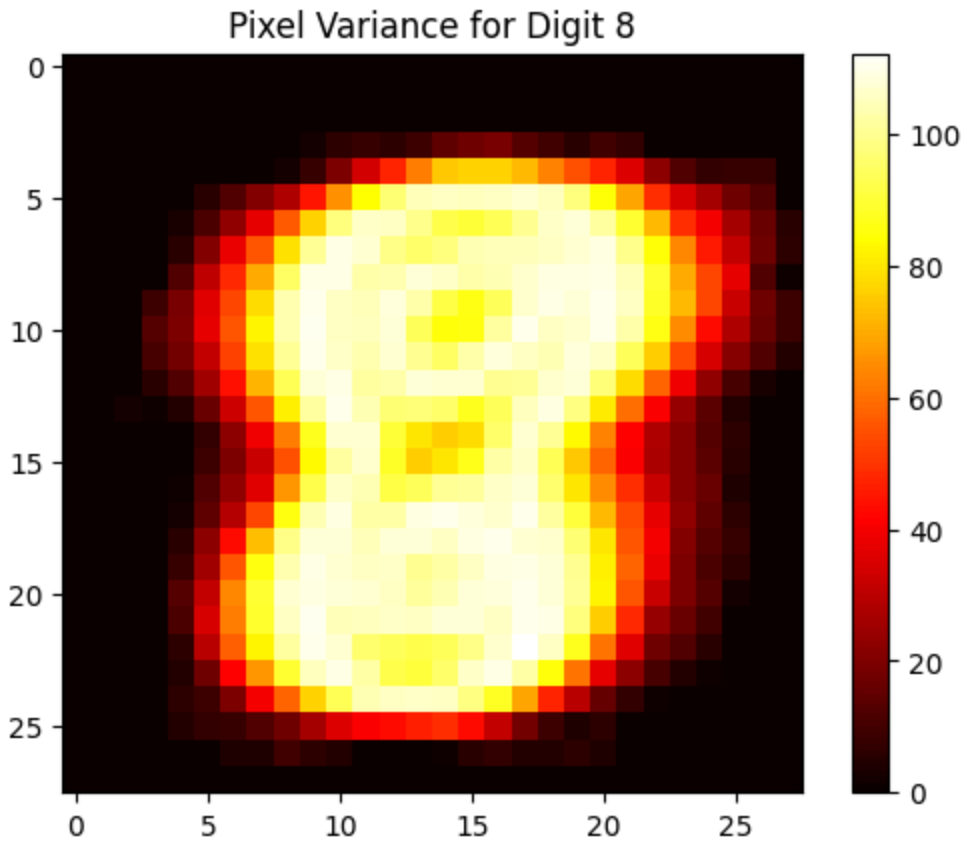


In [67]:
```
plt.figure(figsize=(10,4))
for digit in range(10):
    mean_img = X[y == digit].mean().values.reshape(28,28)
    plt.subplot(2,5,digit+1)
    plt.imshow(mean_img, cmap="gray")
    plt.title(f"Digit {digit}")
    plt.axis("off")
plt.show()
```



In [68]:
```
std_img = X[y == 8].std().values.reshape(28,28)
plt.imshow(std_img, cmap="hot")
plt.title("Pixel Variance for Digit 8")
```

```
plt.colorbar()
plt.show()
```

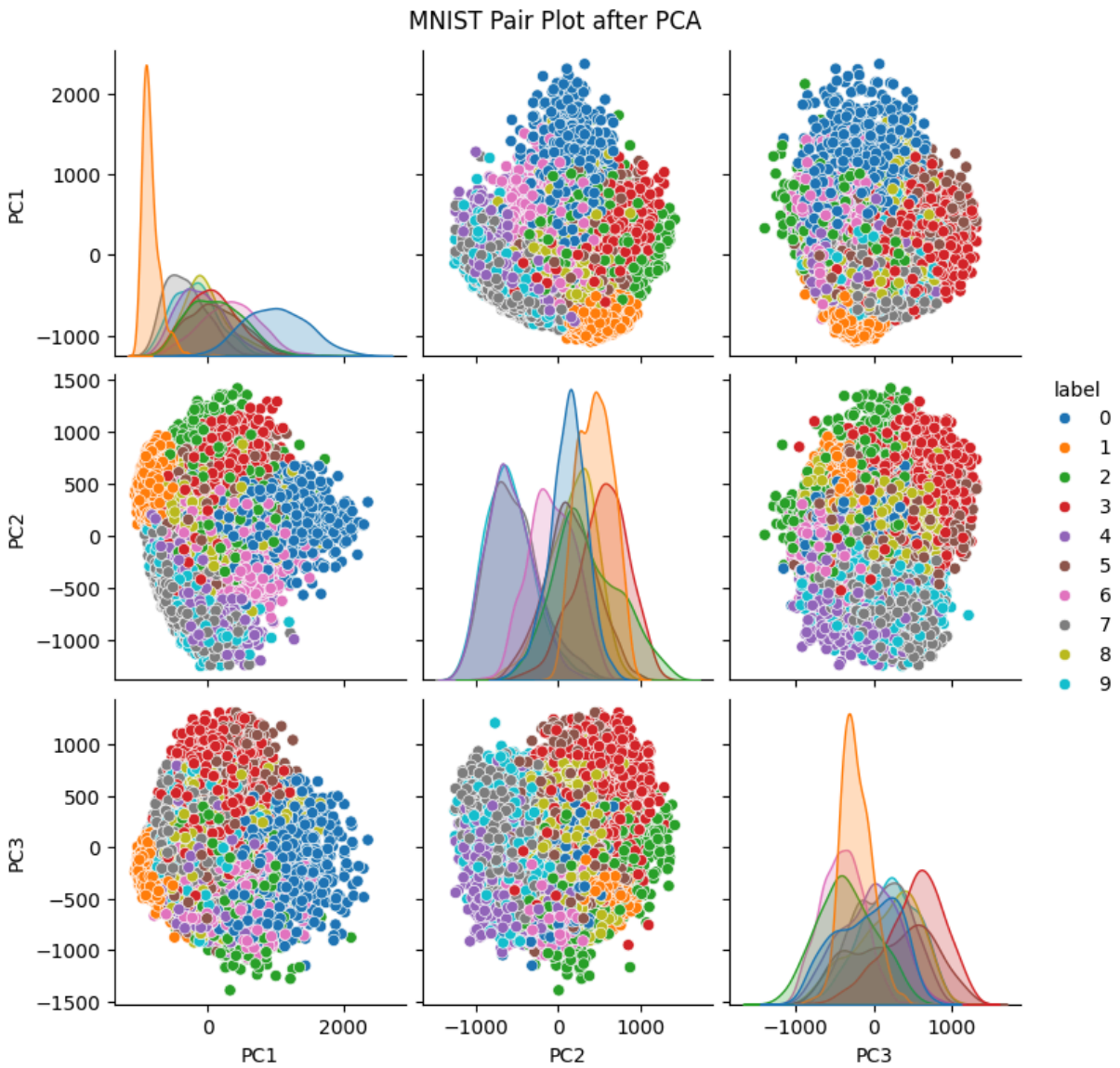


In [69]:
```
X = df.iloc[:, 1:]   # 784 pixel columns
y = df.iloc[:, 0]    # label column
```

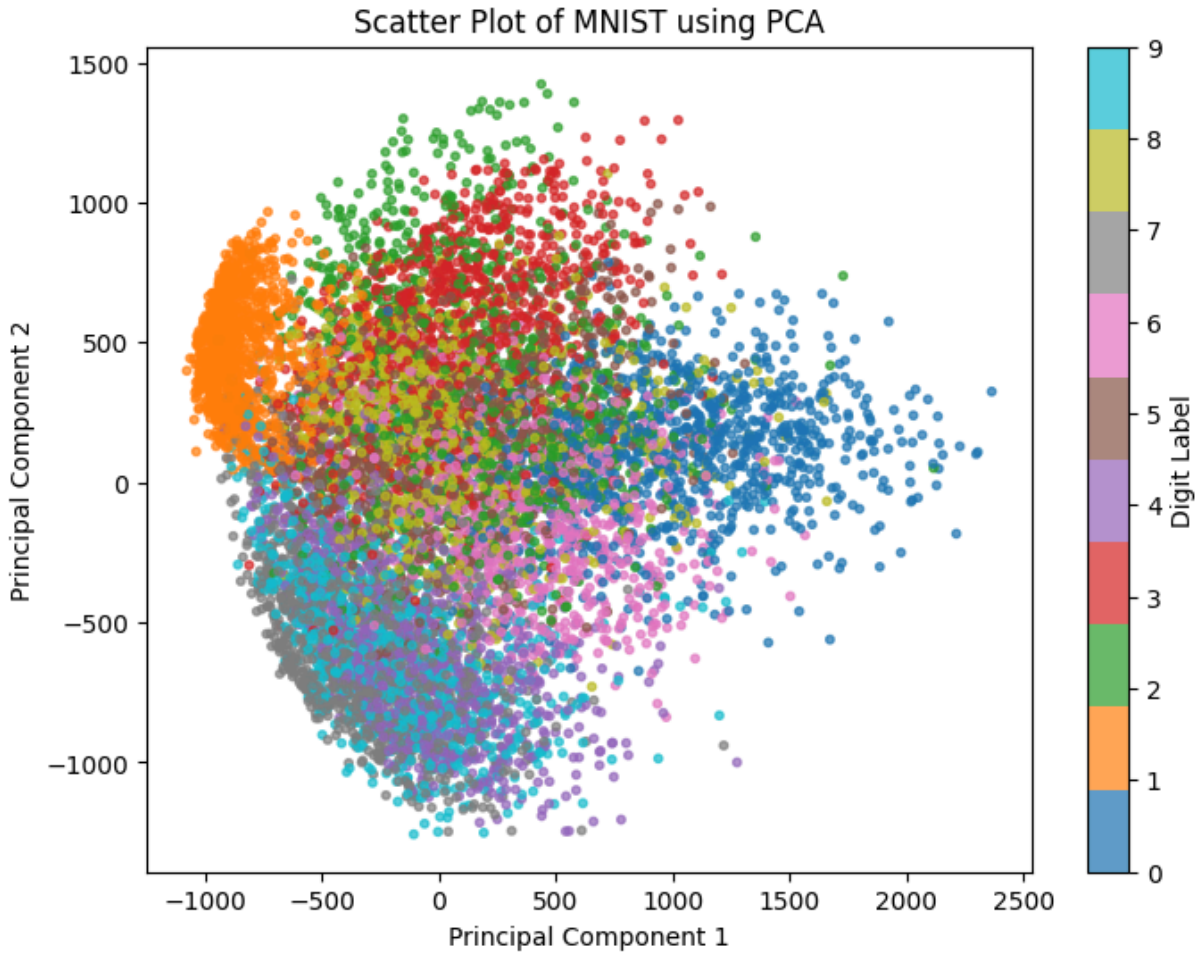
In [70]:
```
pca = PCA(n_components=3)
X_pca = pca.fit_transform(X)
```

In [71]:
```
pca_df = pd.DataFrame(X_pca, columns=["PC1", "PC2", "PC3"])
pca_df["label"] = y
```

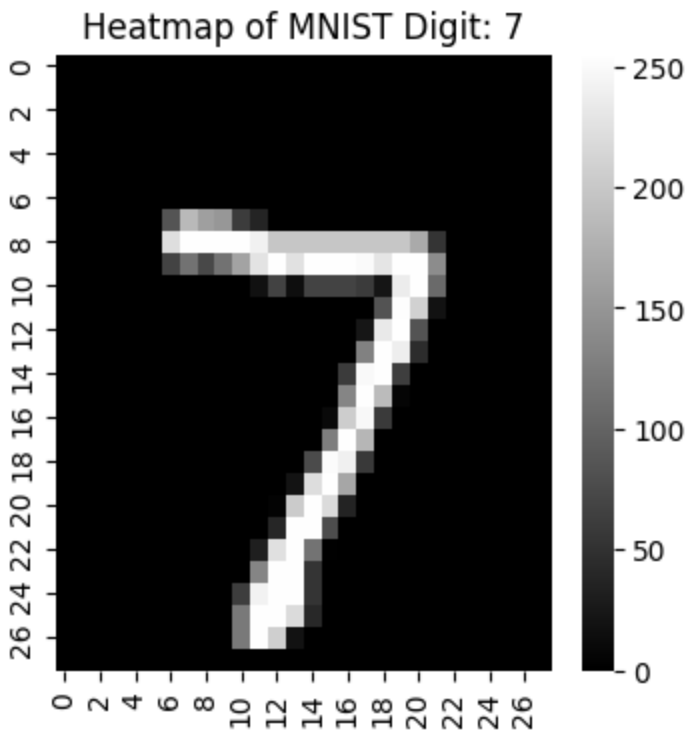
In [72]:
```
sns.pairplot(
    pca_df,
    vars=["PC1", "PC2", "PC3"],
    hue="label",
    palette="tab10",
    diag_kind="kde"
)

plt.suptitle("MNIST Pair Plot after PCA", y=1.02)
plt.show()
```



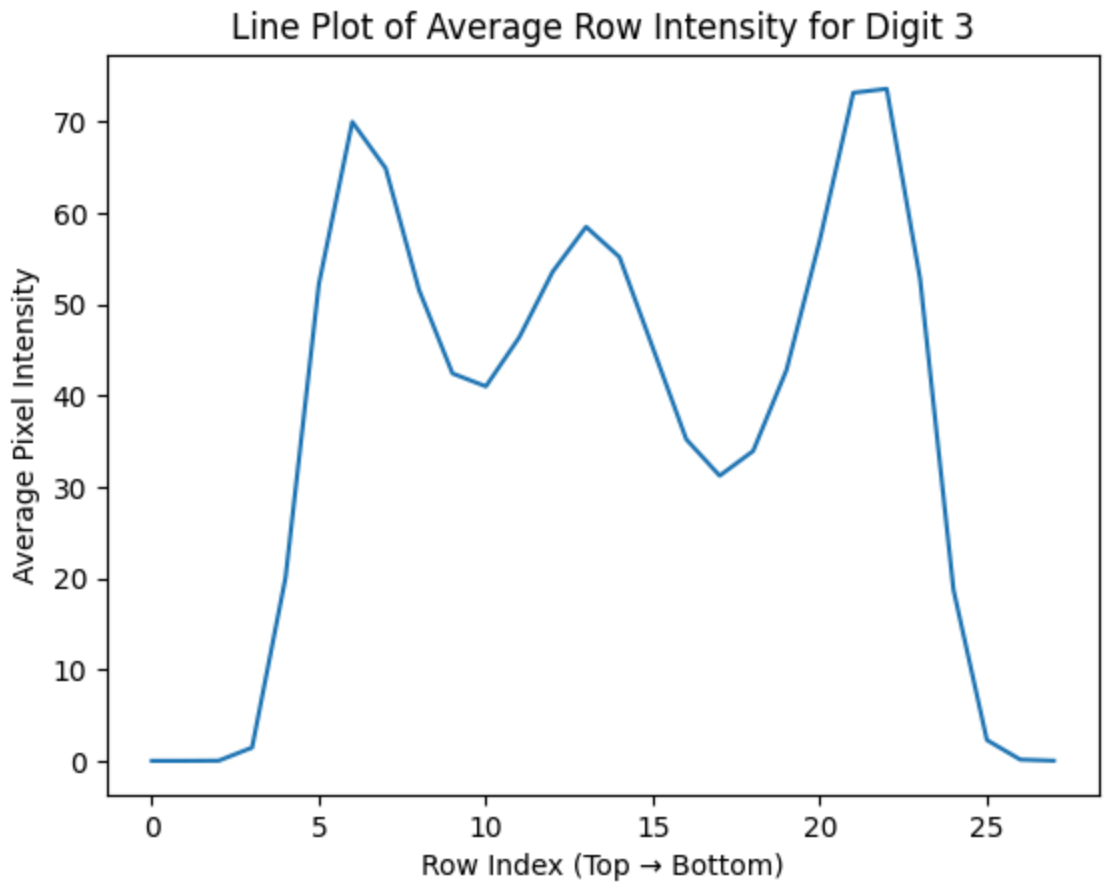
In [75]:
```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

# Scatter plot
plt.figure(figsize=(8,6))
plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=y,
    cmap="tab10",
    s=10,
    alpha=0.7
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Scatter Plot of MNIST using PCA")
plt.colorbar(label="Digit Label")
plt.show()
```



In [78]:
```
# Select one image
img = X.iloc[0].values.reshape(28, 28)

plt.figure(figsize=(4,4))
sns.heatmap(img, cmap="gray", cbar=True)
plt.title(f"Heatmap of MNIST Digit: {y.iloc[0]}")
plt.show()
```



In [79]:
```
digit = 3
digit_imgs = X[y == digit].values.reshape(-1, 28, 28)

# Average intensity per row
avg_row_intensity = digit_imgs.mean(axis=0).mean(axis=1)

plt.plot(avg_row_intensity)
plt.xlabel("Row Index (Top → Bottom)")
plt.ylabel("Average Pixel Intensity")
plt.title(f"Line Plot of Average Row Intensity for Digit {digit}")
plt.show()
```


Line Plot of Average Row Intensity for Digit 3
Average Pixel Intensity
0
10
20
30
40
50
60
70
0
5
10
15
20
25
Row Index (Top → Bottom)

In [ ]: